Memory Products

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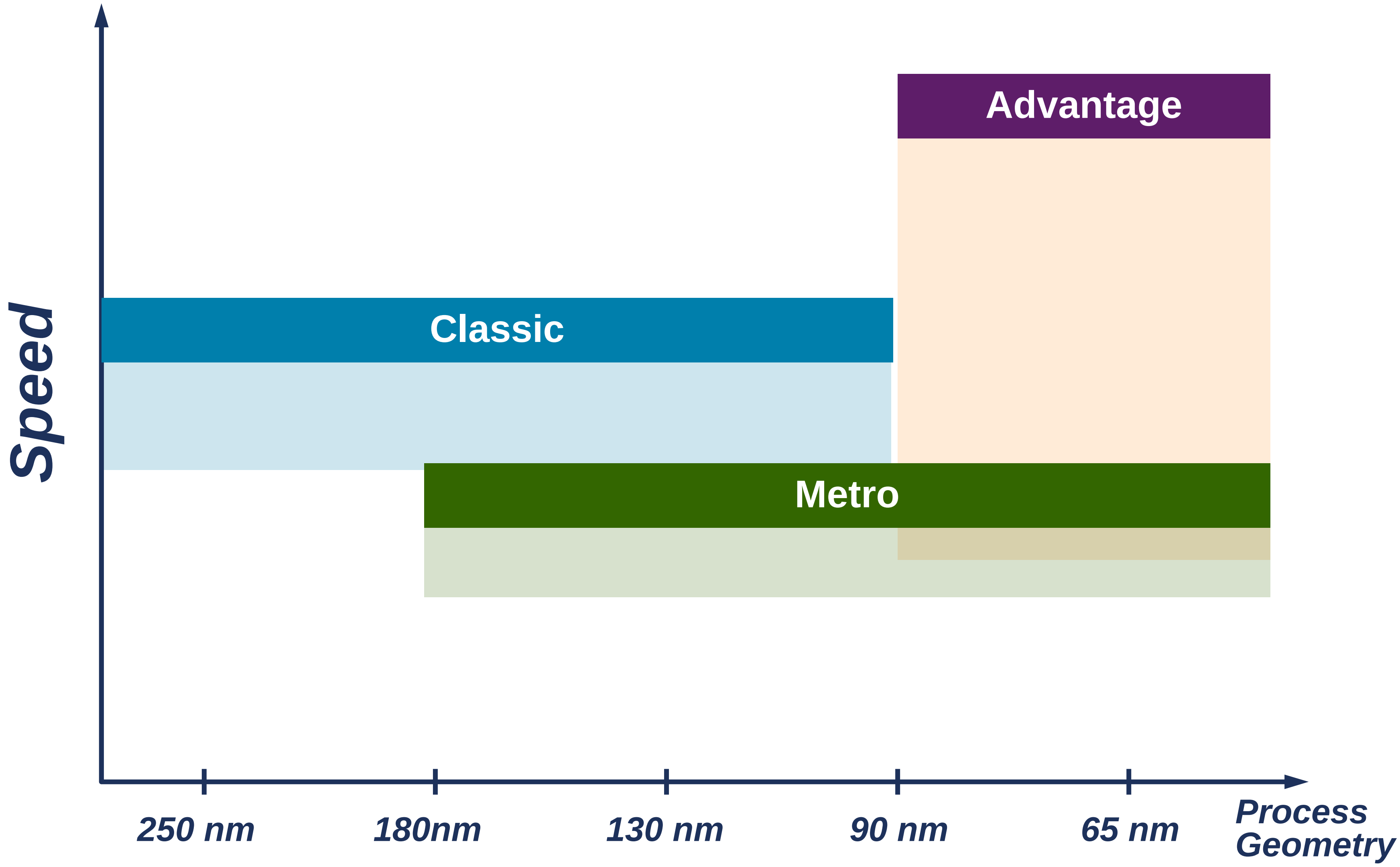
Memory Platform Performance



Comparisons Between Memory Architectures

- Classic values normalized to 1

Parameter	Classic	Advantage	Metro
Speed	1	1.4	0.8
Dynamic Power	1	0.6	0.4
Leakage Power	1	1	0.2
Area	1	1	0.9

Memory Generator Portfolio

Generator	Range	Ports
Single Port SRAM	256 - 512Kbits	1 R/W Port
Dual Port SRAM	128 - 256Kbits	2 R/W Ports
Single Port Register File	64 - 32Kbits	1 R/W Port
Two Port Register File	16 - 16Kbits	1 R Port, 1 W Port
ROM	128 - 2Mbits	1 R Port
Ultra High Density Two Port Register File	16 - 16Kbits	1 R Port, 1 W Port
Multi-Port Register File	16 - 16Kbits	1-9 R Port, 1-2 W Port

Classic Memory Feature Set

- Fully static and synchronous designs
- User configurable mask write options (1 to 36 bits)
- Aspect ratio control for efficient floor planning
- Memory Operation and data retention to 0 Mhz
- Low active power and leakage only standby power
- Flex-Repair™ Redundancy

Classic Memory Physical IP Availability

Foundry	250nm	180 -150nm	130 -110nm
Chartered	✓	✓	✓
IBM	✓	✓	✓
TSMC	✓	✓	✓
UMC	✓	✓	✓
1st Silicon	✓	✓	✓
Dongbu Anam		✓	✓
MagnaChip		✓	✓
SMIC		✓	✓
Silterra		✓	✓
HHNEC		✓	
HeJian		✓	
Tower		✓	✓
Vanguard	✓	✓	

Free Library Program

Metro Low Power Physical IP

METRO™
Low-Power Platform

- **Metro™ Standard Cell**
- **Single-Port/Dual Port SRAM**
- **1-Port/2-Port Register File**
- **ROM Via or Diffusion**
- **GPIO Inline/Staggered**
- **Analog/Mixed Signal PLL, DLL Specialty I/O**
- **Power Management Kit**

- Power-optimized Physical IP products
  - Enables up to 80% lower power than traditional products
  - Dynamic and Static power management
  - Enables Multi-voltage SoC designs
- Compatible with latest releases of EDA vendor ‘power-optimized’ flows
  - Increased design productivity
- Targets battery-powered applications
  - Area efficient
  - Moderate performance
  - Improved yield

Metro Memory – Enhanced Features

- Back Gate Bias and Power Gating Support
- Power Down Modes
  - Standby, Retention, and Shutdown modes
- Low Voltage Functionality and Characterization
  - Enables dynamic power savings
  - Enables multi-voltage island designs
- Integrated Flex-Repair Redundancy for improved yield
  - One column of redundancy for register files
  - Two columns and two rows for SRAMs
- Extra Margin Adjust (EMA)
- Integrated BIST Mux
- Power Rings or ArtiGrid™ Ringless Power Routing
- ECC / Soft Error Repair

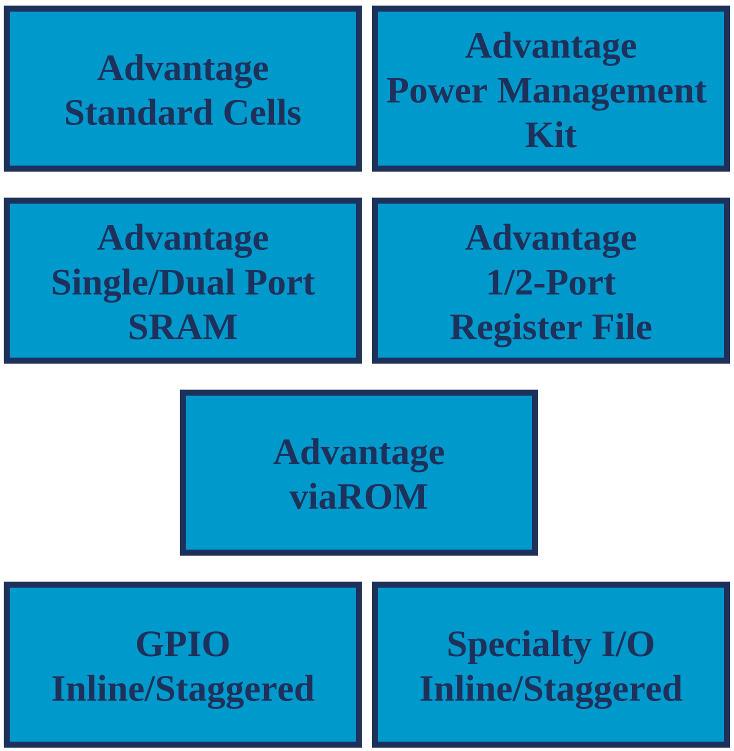
ARM Metro Memory Availability

Foundry	180 -150nm	130 -110nm	90 - 80nm	65nm
Chartered	✓	✓	✓	✓
IBM			✓	✓
TSMC	✓	✓	✓	Announcements Soon
UMC		✓		
1st Silicon	✓			
Dongbu Anam	✓			
MagnaChip	✓			
SMIC	✓	✓	✓	
Silterra	✓			
HHNEC	✓			
HeJian	✓			
Tower	✓			
Vanguard	✓			

Free Library Program

Advantage High Performance Platform

ADVANTAGE™
Performance Platform



- 65nm Enhanced Standard Cells
  - High performance, Higher density
  - Enhanced characterization
  - Optimized routability
- All new memory architecture
  - High performance architecture
  - Optional pipelined outputs
  - Support for a full BIST and BISR DFT solution
- Lower Die Costs
  - Extensive DFT and DFM support for Improved Yield
  - ArtiGrid™ Ring-less Power Routing

Advantage Memory – Enhanced Features

- Ultra High Speed Architecture
  - Capable of 1GHz Performance (@65nm)
  - Optional pipelined outputs (@65nm)
- Optional Power Down Modes (@65nm)
  - Standby, Retention and Shutdown
- Extra Margin Adjustment
  - Multiple choices of internal timing pulse width
  - Allows post silicon timing adjustment to improve yield
- Flex-Repair™ Redundancy
  - One column of integrated redundancy for Register Files
  - Two columns and two rows for SRAMs
- ECC / Soft Error Repair
  - RTL generated for modified Hamming code
- Integrated BIST Test Muxes
- Power Rings Or ArtiGrid™ ringless Power Routing

ARM Advantage Memory Availability

Foundry	180 -150nm	130 -110nm	90 - 80nm	65nm
Chartered			✓	✓
IBM			✓	✓
TSMC			✓	*Announcements Soon*
UMC			✓	
1st Silicon				
Dongbu Anam				
MagnaChip				
SMIC			✓	
Silterra				
HHNEC				
HeJian				
Tower				
Vanguard				

Free Library Program

Specialty Memory Generators

- High Density Register Files
  - High Density Two Port Register File
  - Ultra High Density Two Port Register File
- Multi-Port Register Files
  - Up to 11 ports
  - Choice of synchronous or asynchronous access

High Density Register Files

- High Density 2-Port Register File
  - 20% smaller than mainstream 2-port register file
  - 15% lower performance than mainstream 2-port register file
  - Asynchronous clocks
  - Proven in volume customer silicon
  - Available in multiple foundries at 130nm and 90nm
- Ultra High Density 2-port Register File
  - 30% smaller than mainstream 2-port register file
  - 20% lower performance than mainstream 2-port register file
  - Synchronous clocks
  - Silicon proven
  - Available in TSMC 90nm G and GT

Multi-Port Memory

- Multi-port register files
  - Up to 2 write ports
  - Application note or GDS2

- Comparison with competitor

TSMC 130G	3R2W 64x64		3R1W 64x64	
	Area (mm^2)	Access Time (ns)	Area (mm^2)	Access Time (ns)
Competitor	0.215	1.94	0.164	1.72
ARM	0.195	1.55	0.154	1.55
Improvement	9.3%	20.1%	6.1%	9.88%

Memory Availability – 90nm to 150nm

PROCESS	90nm			130nm								150nm
MEMORY GENERATORS	UMC	IBM	Chartered	UMC	Chartered			IBM		SMIC	Tower	SMIC
Advantage SP-SRAM	90SP	9SF	CSM 9SF									
Advantage DP-SRAM	90SP	9SF	CSM 9SF									
Advantage SP-RF	90SP	9SF	CSM 9SF									
Advantage 2P-RF	90SP	9SF	CSM 9SF									
Metro SP-SRAM (Low Power)		9FLP	9FLP	L130E	130N	130LP						
Metro DP-SRAM (Low Power)		9FLP	9FLP	L130E	130N	130LP						
Metro SP-RF (Low Power)		9FLP	9FLP	L130E	130N	130LP						
Metro 2P-RF (Low Power)		9FLP	9FLP	L130E	130N	130LP						
Metro vROM (Low Power)		9FLP	9FLP	L130E	130N	130LP						
Classic SP-SRAM-HS/HD*					130N	130LP	130HP-HVT	8RF	8SFG	130G	13SL	150G
Classic SP-SRAM-HS/HD* w/ Flex Rpr					130N	130LP	130HP-HVT	8RF	8SFG	130G	13SL	150G
FuseBox	90SP			L130E	130N	130LP	130HP-HVT			130G	13SL	150G
Classic DP-SRAM-HS/HD*					130N	130LP	130HP-HVT	8RF	8SFG	130G	13SL	150G
Classic SP-RF-HS/HD*					130N	130LP	130HP-HVT	8RF	8SFG	130G	13SL	150G
Classic 2P-RF-HS/HD*					130N	130LP	130HP-HVT	8RF	8SFG	130G	13SL	150G
Classic 2P-RF-HD					130N	130LP	130HP-HVT				13SL	
Classic dROM-HS/HD*					130N	130LP	130HP-HVT			130G	13SL	
Classic vROM-HS/HD*					130N	130LP	130HP-HVT	8RF	8SFG	130G	13SL	150G

Explanation	
Free library program (available now)	
Free library program (in development)	
Application-specific - OTS (available now)	
Application-specific - OTS (in development)	
Not Available	

Memory Availability – 180nm

PROCESS	180nm												
MEMORY GENERATORS	HeJian	HHNEC	Grace	Dongbu 180nm		Chartered 180nm				First Silicon 180nm			
Metro SP-SRAM (Low Power)	HEJ 018	180G	180G	180BB		180 IB		180FC	180N			180	180FC
Metro SP-SRAM-ULP (Ultra Low Pwr)	HEJ 018	180G	180G	180BB		180 IB		180FC	180N			180	180FC
Metro DP-SRAM (Low Power)	HEJ 018	180G	180G	180BB		180 IB		180FC	180N			180	180FC
Classic SP-SRAM-HS/HD*	HEJ 018	180G	180G	180BB	180SA	180 IB	180 CB	180FC	180N	180LP	180 FCLP	180	180FC
Classic SP-SRAM-HD (High Density)	HEJ 018							180FC					
Classic DP-SRAM-HS/HD*	HEJ 018	180G	180G	180BB	180SA	180 IB	180 CB		180N	180LP	180 FCLP	180	180FC
Classic SP-RF-HS/HD*		180G	180G	180BB		180 IB	180 CB					180	180FC
Classic 2P-RF-HS/HD*		180G	180G	180BB		180 IB	180 CB					180	180FC
Classic vROM-HS/HD*		180G	180G	180BB		180 IB	180 CB		180N			180	180FC
Classic dROM-HS/HD*		180G	180G	180BB		180 IB	180 CB		180N			180	180FC
Fusebox		180G	180G	180BB		180 IB	180 CB		180N			180	180FC

Explanation	
Free library program (available now)	
Free library program (in development)	
Application-specific - OTS (available now)	
Application-specific - OTS (in development)	
Not Available	

Memory Availability – 180 and 250nm

PROCESS	250nm				180nm									
MEMORY GENERATORS	First Silicon		UMC	Vanguard	UMC			Tower	Silterra		SMIC	MagnaChip	IBM	
Metro SP-SRAM (Low Power)								TS18SL		180G	180G	180G		
Metro SP-SRAM-ULP (Ultra Low Pwr)								TS18SL		180G	180G	180G		
Metro DP-SRAM (Low Power)								TS18SL		180G	180G	180G		
Classic SP-SRAM-HS/HD*	250	250FC			180LL	180 GII	L180	TS18SL	180LP	180G	180G	180G	7SF	7RF
Classic SP-SRAM-HD (High Density)				VT025LG	180LL	180 GII								
Classic SP-SRAM-HS (High Speed)			L250											
Classic DP-SRAM-HS/HD*	250	250FC				180 GII							7SF	7RF
Classic DP-SRAM-HD								TS18SL	180LP	180G	180G	180G		
Classic DP-SRAM-HS			L250											
Classic SP-RF-HS/HD*								TS18SL	180LP	180G	180G	180G	7SF	7RF
Classic 2P-RF-HS/HD*	250	250FC						TS18SL	180LP	180G	180G	180G	7SF	7RF
Classic 2P-RF-HS												180G		
Classic dROM-HS/HD*	250	250FC						TS18SL	180LP	180G	180G	180G		
Classic vROM-HS/HD*								TS18SL	180LP	180G	180G	180G	7SF	7RF

Explanation	
Free library program (available now)	
Free library program (in development)	
Application-specific - OTS (available now)	
Application-specific - OTS (in development)	
Not Available	

TSMC Processes And Memory Availability

PRODUCTS	CLN90 G StdV$_T$	CLN90 LP StdV$_T$	CLN90 GT StdV$_T$	CL011 LVP	CL011 G	CL013 G StdV$_T$	CL013 LV FSG StdV$_T$	CL013 LV FSG StdV$_T$ OD	CL013 LV LK StdV$_T$	CL013 LV LK StdV$_T$ OD	CL013 LP StdV$_T$	CL015 G StdV$_T$	CL015 LV StdV$_T$	CL018 G	CL018 LV	CL018 LP	CL025 G
MEMORY GENERATORS																	
Advantage SP-SRAM			NOW														
Advantage DP-SRAM			NOW														
Advantage SP-RF			NOW														
Advantage 2P-RF			NOW														
Classic SP-SRAM-HS/HD*				NOW													
Classic SP-SRAM-HS/HD* w/ Flex Rpr				NOW				NOW					NOW [1]				
FuseBox			NOW				NOW	NOW					NOW				
Classic SP-SRAM-HD (High Density)																	
Classic SP-SRAM-HS (High Speed)																	
Metro SP-SRAM-LP														NOW [1]			
Metro SP-SRAM-ULP														NOW			
Metro DP-SRAM LP														NOW			
Metro RF-SP																	
Metro RF-2P																	
Metro vROM																	
Classic SP-RF-HS/HD*				NOW													
Classic DP-SRAM-HS/HD*								NOW [1]									
Classic DP-SRAM-HD																	
Classic 2P-RF-HS/HD*				NOW													
Classic 2P-RF-HS																	
Classic 2P-RF-HD				NOW			NOW	NOW									
Classic 2P-RF-UHD			NOW														
Classic dROM-HS/HD*				NOW				NOW									NOW
Advantage dROM-HD			NOW														
Classic vROM-HS/HD*				NOW										NOW			NOW

Performance Plays

Power Plays

Density Plays

* optimized for speed and density

Free Library Program products

Direct Channel products NOW

Memory Product Roadmap

January 8, 2006

CY 2006 Q1	CY 2006 Q2	CY 2006 Q3	CY 2006 Q4	CY 2007 Q1
UMC L130e **Metro** SP-SRAM DP-SRAM SP-Reg File 2P-RegFile viaROM	**TSMC 130G** **Metro** SP-SRAM	**TSMC 130G** **Metro** SP-RegFile 2P-RegFile		
TSMC 90G **Advantage** High Speed Cache Memory For 1176 Cores		**TSMC 90G** **Metro** SP-SRAM SP-Reg File	**TSMC 90G** **Metro** 2P-RegFile	**TSMC 90G** **Metro** DP-SRAM viaROM
TSMC 90LP **Advantage** SP-SRAM SP-Reg File		**SMIC 90LL** **Metro** SP-SRAM DP-SRAM SP-Reg File 2P-RegFile viaROM		**45nm** **Advantage** SP-SRAM DP-SRAM SP-RegFile 2P-RegFile
SMIC 90G **Advantage** SP-SRAM DP-SRAM SP-Reg File 2P-RegFile viaROM	**CMOS10SF** **Advantage** SP-SRAM DP-SRAM SP-Reg File 2P-RegFile viaROM	**TSMC 65G** **Advantage** SP-SRAM DP-SRAM SP-Reg File 2P-RegFile 2P-Reg File UHD	**TSMC 65G** **Advantage** viaROM	**TSMC 65G** **Metro** SP-SRAM DP-SRAM SP-RegFile 2P-RegFile
CMOS10LP **Metro** SP-SRAM DP-SRAM SP-Reg File 2P-RegFile viaROM	**TSMC 65G** **Advantage** 2P-RegFile UHD			

Advantage = High Performance Platform

Metro = Low Power Platform

Right edge of box = Production Release

Free Library Program (yellow boxes)

User Licensable Off-The-Shelf (green boxes)